

GENERAL SPECIFICATION G5.38

Low Voltage AC Induction Motors

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G5.38 LOW VOLTAGE AC INDUCTION MOTORS

G5.38.1 General

(a) Introduction

- (i) Perform all work; design, supply, deliver and install all materials and equipment, and installation methods in accordance with, but not limited to, the latest applicable rules, regulations, requirements, national and international codes, and specifications as listed in General Specification G5.2- Standards, Codes and Regulations.

(b) References

- (i) The following is a list of standards which may be referenced in this Section:

1) International Electrotechnical Commission (IEC)

Standard	Subject
IEC 60034	Rotating Electric Machines.
IEC 60072	Dimensions and Output Series for Rotating Electrical Machines.
IEC 60079	Explosive atmospheres.

2) Spring Singapore (SS)

Standard	Subject
CP 5	Code of Practice for Electrical Installation.
SS 551	Code of Practice for Earthing.
SS 555	Code of Practice against lightning protection

- (ii) Conflicts, if any, that may exist between the above reference standards, codes and regulations will be resolved at the discretion of the S.O..

(c) Abbreviations

Abbreviations	Subject
CISD-TEFC	Chemical industry, severe-duty enclosure
DIP	Dust-Ignition-Proof Enclosure
EXP	Explosion-Proof Enclosure
ODP	Open Drip-Proof Enclosure
TEFC	Totally Enclosed, Fan Cooled Enclosure
TENV	Totally Enclosed, Non-Ventilated Enclosure

Abbreviations	Subject
WPI	Weather Protected Enclosure, Type I
WPII	Weather Protected Enclosure, Type II

- (i) Motor Nameplate Rating: That rating after any derating required to allow for extra heating caused by the harmonic content in the voltage applied to the motor by its controller.
- (ii) Inverter Duty Motor: Motor meeting or exceeding all applicable requirements of IEC.
- (d) Submittals
 - (i) Shop Drawings, and Technical Data: As a minimum, submit the following items:
 - 1) Descriptive information;
 - 2) Nameplate data in accordance with IEC 60034;
 - 3) Motor manufacturer address and contact details;
 - 4) Motor name and Specification number of driven equipment;
 - 5) Frame size;
 - 6) Additional Information:
 - Service Factor;
 - Locked rotor current;
 - No load current;
 - Safe stall time for motors 100 kilowatts and larger;
 - Multispeed load classification (e.g., variable torque);
 - Variable frequency drive motor load classification (e.g., variable torque) and minimum allowable motor speed for that load classification;
 - Maximum number of successive cold and hot starts;
 - Code letter for locked rotor. Locked rotor current in % of full load current;
 - Motor thermal performance- hot and cold start curves;
 - Guaranteed minimum full load efficiency and efficiencies at ½ load and ¾ load;
 - Design Standard;
 - Starting Method (DOL, Star/Delta , soft Starter or VFD) and number of starts per hour;
 - Space heater voltage and watts;
 - Power factor at ½, ¾ and full load;

- Locked rotor, pull up, and breakdown torque;
 - Full load load-torque and super imposed with the curve of the driven equipment;
 - Motor termination drawings with RTD wiring at site. Motor and pump plinth as-built drawings;
 - Motor main and auxiliary terminal boxes dimensions, location on the motor and knockout details;
 - Rotor inertia; and
 - Manufacturer shall certify that the motors are driven by VFDs are inverter duty motors and shall confirm that the motor is suitable for the particular driven equipment and shall include the minimum speed in which the motor can operate.
- 7) Enclosure type and mounting (e.g., horizontal, vertical). State clearly whether it is for hazardous or non-hazardous locations that confirms to IEC Zone 0, Zone 1 and Zone 2 conform to IECEx and have an applied IECEx listing mark if applicable and as installed at site;
 - 8) Type test certificate All assembly clearances and tolerances, including but not limited to the diametrical bearing clearances, bearing housing crush, air gap and coupling interference fit to shaft shall be submitted;
 - 9) All assembly radial and axial runout tolerances shall be submitted. This requirement includes but not limited to shaft runout at various lateral locations, plus rotor face runout if applicable;
 - 10) Motor type or model, Dimensional drawing and total weight;
 - 11) Conduit box dimensions and usable volume;
 - 12) Bearing type;
 - 13) Bearing lubrication;
 - 14) Bearing life;
 - 15) Description, rating, and wiring diagram of motor thermal protection;
 - 16) Motor sound power level in accordance with IEC 60034;
 - 17) Maximum power, in kilowatts, number of phases, frequency, rated speed, number of poles;
 - 18) Full load current at rated output, rated speed and rated voltage, required by the equipment driven by the motor.;
 - 19) Winding insulation class and temperature rise;
 - 20) Factory Type Test reports;
 - 21) FAT reports;
 - 22) Manufacturer's Certificate of Proper Installation; and
 - 23) Operation and Maintenance Manual.

G5.38.2 Products

- (a) General
 - (i) For multiple units of the same type of equipment, provide identical motors and accessories of a single manufacturer.
 - (ii) In order to obtain single source responsibility, utilise a single manufacturer to provide a drive motor, its driven equipment, and specified motor accessories.
 - (iii) Meet requirements of IEC 60034 and IEC 60072.
 - (iv) Frame assignments in accordance with IEC 60072.
 - (v) Provide motors for hazardous (classified) locations that conform to IEC Zone 0, Zone 1 and Zone 2, conform to IECEx and have an applied IECEx listing mark.
 - (vi) Motors shall be specifically designed for the use and conditions intended, with IEC design letter classification to fit the application.
 - (vii) Lifting lugs on all motors weighing 20 kg or more.
 - (viii) Operating Conditions
 - 1) Maximum ambient temperature not greater than 40 degrees C;
 - 2) Elevation: <1000 m above Mean Sea Level;
 - 3) Motors shall be suitable for operating conditions without any reduction being required in the nameplate rated power or exceeding the rated temperature rise; and
 - 4) Overspeed in either direction in accordance with IEC 60034.
- (b) Kilowatt Rating
 - (i) As designated in motor-driven equipment specifications;
 - (ii) Constant Speed Applications: Brake kilowatt of the driven equipment at any operating condition not to exceed motor nameplate kilowatt rating, excluding service factor; and
 - (iii) Variable Frequency, Adjustable Speed Applications (Inverter Duty Motors): Driven equipment brake kilowatt at any operating condition not to exceed motor nameplate kilowatt rating, excluding service factor.
- (c) Service Factor
 - (i) Inverter-duty Motors: 1.0 at rated ambient temperature, unless otherwise specified; and
 - (ii) Other Motors: 1.15 minimum at rated ambient temperature, unless otherwise specified.
- (d) Voltage And Frequency Rating
 - (i) System Frequency: 50-Hz;

- (ii) Voltage Rating: Unless otherwise specified in motor-driven equipment specifications:

Voltage Rating		
Size	Voltage	Phase
0.75 kW and smaller	230	Single Phase
Larger than 0.75 kW	400	Three Phase

- (iii) Suitable for full voltage starting;
- (iv) Motors rated 15 kilowatts and above shall be suitable for reduced voltage starting; and
- (v) Suitable for accelerating the connected load with supply voltage at motor starter supply terminals dipping to 90 percent of motor rated voltage.
- (e) Efficiency and Power Factor
- (i) Efficiency
- 1) IE3 efficiency for constant speed motors, 0.75 kW and larger;
 - 2) IE3 efficiency for VFD driven IEC motors except single-phase, under 0.75 kilowatt, multispeed, and short-time rated, or motors driving gates, valves, elevators, cranes, trolleys, and hoists; and
 - 3) Efficiency: Tested in accordance with IEC 60034. Guaranteed minimum efficiency at full load in accordance with IEC or as indicated in motor driven equipment specifications.
- (ii) Power Factor
- 1) Guaranteed minimum power factor at full load shall be manufacturer's standard or as specified in motor catalogue for motors rated up to 7.5 kW; and
 - 2) Guaranteed minimum power factor at full load shall be 0.86 for motor rated above 7.5 kW.
- (f) Locked Rotor Ratings
- (i) Locked rotor kVA Code G or lower if motor rating not covered by IEC 60034; and
- (ii) Safe Stall Time: 12 seconds or greater.
- (g) Insulation Systems
- (i) Single-Phase, Fractional Power Motors: Manufacturer's standard winding insulation system;
- (ii) Tropicalisation: Insulation systems shall be suitable for tropical applications with a minimum of three dip and bake applications; and
- (iii) Three-Phase Motors, Unless Otherwise Specified in Motor-Driven Equipment Specifications: Class F with Class B temperature rise at nameplate rating and designated operating conditions.

- (h) Enclosures
 - (i) All enclosures to conform to IEC 60034-5;
 - (ii) TEFC (IP 55) and CISC-TEFC (IP 55): Supply with a drain hole with porous drain/weather plug;
 - (iii) Explosion-Proof (EXP)
 - 1) TEFC listed to meet IEC 60079 requirements for IEC Zone 0, Zone 1 and Zone 2 hazardous locations;
 - 2) Drain holes with drain and breather fittings;
 - 3) Integral thermostat or thermistor opening on excessive motor temperature in accordance with IEC standard; and
 - 4) Thermostat / thermistor leads to terminate in a terminal box separate from main terminal box.
 - (iv) Dust-Ignition-Proof (DIP)
 - 1) TEFC listed to be met IEC 60079 requirements for hazardous locations;
 - 2) Integral thermostat or thermistor opening on excessive motor temperature in accordance with UL IEC standard; and
 - 3) Thermostat / thermistor leads to terminate in a terminal box separate from main terminal box.
 - (v) Submersible: In accordance with article Special Motors; and
 - (vi) Chemical Industry, Severe-Duty (CISC-TEFC): In accordance with article Special Motors.
- (i) Terminal (Conduit) Boxes
 - (i) Oversize main terminal boxes for all motors;
 - (ii) Diagonally split, rotatable to each of four 90-degree positions. Threaded hubs for conduit attachment;
 - (iii) Provide gaskets between box halves and provide seal between box and motor frame;
 - (iv) Minimum usable volume in percentage shall be shown as below table:

Voltages	Kilowatts	Percentage
Below 600	7.5 through 75	500
Below 600	82 through 224	275
Below 600	Greater than 224	225

- (v) Terminal for connection of equipment earthing wire in each terminal box; and
- (vi) Terminal boxes shall be threaded to accept cable glands with metric threads.

- (j) Bearings and Lubrication
 - (i) Horizontal Motors
 - 1) One kilowatt and Smaller: Permanently lubricated and sealed ball bearings, or regreasable ball bearings in labyrinth sealed end bells with removable grease relief plugs;
 - 2) Greater Than 1 kilowatt Through 300 kilowatts: Regreasable ball bearings in labyrinth sealed end bells with removable grease relief plugs;
 - 3) Above 300 Kilowatts: Split sleeve, oil insulated bearings; and
 - 4) Minimum 100,000 hours L-10 bearing life for ball and roller bearings as defined in ABMA 9 and 11.
 - (ii) Vertical Motors
 - 1) Thrust Bearings:
 - Manufacturer's standard lubrication; and
 - Minimum 50,000 hours L-10 life.
 - 2) Guide Bearings:
 - Manufacturer's standard bearing type;
 - Manufacturer's standard lubrication; and
 - Minimum 100,000 hours L-10 bearing life.
 - (iii) Regreasable Antifriction Bearings
 - 1) Readily accessible, grease injection fittings;
 - 2) Readily accessible, removable grease relief plugs;
 - 3) Greasing frequency shall be submitted.; and
 - 4) Type and model of recommended greasing guns shall be submitted.
 - (iv) Oil Lubrication Systems
 - 1) Oil reservoirs with sight level gauge;
 - 2) Oil fill and drain openings with opening plugs; and
 - 3) Provisions for necessary oil circulation and cooling.
 - (v) Bearing Isolation: Equip motors greater than 37 kW to be powered by variable frequency drive units with electrically isolated bearings on both sides (i.e. DE and NDE) to prevent stray current damage.
- (k) Noise
 - (i) Measured in accordance with IEC 60034-9; and
 - (ii) Motors controlled by variable frequency drive systems shall not exceed sound levels of 3 dBA higher as per IEC Standard.

- (l) Balance and Vibration Control
 - (i) In accordance with IEC 60034-14; and
 - (ii) Vibration sensors and monitoring system shall be provided as specified in the motor-driven equipment specifications.
- (m) Equipment Finish
 - (i) External Finish: Prime and finish coat manufacturer's standard and suitable for H₂S corrosive environment based on the manufacturer's recommendation suitable for actual site condition; and
 - (ii) Internal Finish: Bore and end turns coated with clear polyester or epoxy varnish, tropicalised.
- (n) Special Features And Accessories
 - (i) Screen Over Air Openings: Stainless steel on all motors with openings. Attach with stainless steel screws; and
 - (ii) Winding Thermal Protection:
 - 1) Thermostats:
 - Provide in motors for constant and variable speed application 5.5 through 74 kilowatts, unless otherwise specified;
 - Bi-metal disk or rod type thermostats embedded in stator windings;
 - Automatic reset contacts rated 24 volts AC, 5 amperes minimum, opening on excessive temperature (Manual reset shall be provided at motor controller.); and
 - Leads extending to separate terminal box for all motors.
 - 2) Thermistors:
 - Provide in motors for constant and variable speed application 75 kilowatts and larger, unless otherwise indicated;
 - Thermistor embedded in each stator phase winding before winding dip and bake process;
 - In intimate contact with winding conductors;
 - Epoxy-potted, solid state thermistor control module mounted in IP 66 box on motor by motor manufacturer;
 - Individual thermistor circuits factory-wired to control module;
 - Solid-state thermistor control module rated for 24 volts AC power supply for mounting in motor controller cabinet;
 - Control module automatically reset contact for external use rated 24 volts AC, 5 amperes minimum, opening on abnormally high winding temperature. Manual reset shall be provided at motor controller; and
 - Control module contacts shall not change state on loss of control power.

- 3) Resistance Temperature Detector:
 - Motors for constant speed and adjustable speed application 200 kW and larger;
 - 100-ohm platinum, three-wire, precision resistors with calibrated resistance-temperature characteristics;
 - Six (two each phase) positioned to detect highest winding temperature and located between coil sides in stator slots;
 - Compatible with monitoring instrumentation provided with motor controller and with adjustable speed control equipment; and
 - Leads brought to separate motor terminal box.
- (iii) Bearing Temperature Protection: As specified in the motor-driven equipment specifications
- (iv) Vibration Detection and Monitoring: As specified in the motor driven equipment specifications.
- (v) Space Heaters
 - 1) Provide winding space heaters with leads wired out to separate terminal box;
 - 2) Unless shown otherwise, heater shall be suitable for 230 volts AC supply, with wattage suitable for frame size;
 - 3) Space heaters shall be provided for motors rated 2.2 kW and above, located indoor, and
 - 4) Space heaters shall be provided for all motors, located outdoor.
- (vi) Nameplates
 - 1) Raised or stamped letters on stainless steel or aluminium with manufacturer serial number, codes etc,
 - 2) Display all motor data required by IEC 60034-1; and
 - 3) Premium efficiency motor nameplates to also display nominal efficiency, full load power factor, and maximum allowable kVAR for power factor correction capacitors.
- (o) Anchor Bolts: Provide meeting manufacturer's recommendations and of sufficient size and number for specified seismic condition.
- (p) Special Motors
 - (i) Requirements in this article take precedence over conflicting features specified elsewhere in this section.
 - (ii) Inverter Duty Motors:
 - 1) Motors supplied power by a variable frequency drive shall be rated for inverter duty application by the manufacturer;
 - 2) Motor shall be suitable for operation over the entire speed range indicated in the equipment specification;

- 3) Provide forced ventilation where speed ratio is greater than published range for motor being installed;
 - 4) When installed in hazardous classified location shall be identified as acceptable for variable speed when used for such locations; and
 - 5) Shaft Grounding Devices: Motors larger than 75 kW shall be provided with shaft grounding brush or conductive micro fibre shaft grounding ring. Shaft grounding sleeve shall be solidly bonded to grounded motor frame per manufacturer's recommendations.
- (iii) Chemical Industry, Severe-Duty (CISD-TEFC)
- 1) In accordance with IEC;
 - 2) TEFC (IP 55 and IC 41) in accordance with IEC 60034;
 - 3) Suitable for indoor or outdoor installation in severe-duty applications including high humidity, chemical (corrosive), dirty, or salty atmospheres;
 - 4) Motor Frame, End Shields, Terminal Box(es), and Fan Cover: Cast iron;
 - 5) Ventilating Fan: Corrosion-resistant, non-sparking, external;
 - 6) Drain and Breather Fittings: Stainless steel;
 - 7) Nameplate: Stainless steel;
 - 8) Gaskets between terminal box halves and terminal box and motor frame;
 - 9) Extra slinger on rotor shaft to prevent moisture seepage along shaft into motor;
 - 10) Double shielded bearings;
 - 11) 125,000 hours minimum L-10 bearing life for direct-connected loads;
 - 12) External Finish: Double-coated epoxy enamel;
 - 13) Coated rotor and stator air gap surfaces;
 - 14) Insulation System, Windings, and Connections:
 - Class F insulation, Class B rise or better at 1.0 service factor; and
 - Multiple dips and bakes of non-hygroscopic polyester varnish.
 - 15) Service Factor:
 - At 40 Degrees C Ambient: 1.15; and
 - At 65 Degrees C Ambient: 1.00.
 - 16) Safe Stall Time Without Injurious Heating: 20 seconds minimum.
- (iv) Severe-Duty Explosion-Proof: Meet requirements for EXP enclosures and CISD-TEFC motors.
- (v) Severe-Duty, Dust-Ignition-Proof: Meet requirements for DIP enclosures and CISD-TEFC motors.
- (vi) Multispeed: Meet requirements for speeds, number of windings, and load torque classification indicated in the motor-driven equipment specifications.

- (vii) Submersible Pump Motors: Submersible motors shall be closely coupled to the pump to form integral unit. In addition to the foregoing requirement, submersible motors shall comply with the following additional requirements.

- 1) Motor capable of running dry continuously;
- 2) Enclosure:
 - IP 68, Hermetically sealed, watertight, for continuous submergence up to 20 metres depth; and
 - Seals: Tandem mechanical.
- 3) Bearing and Lubrication:
 - Permanently sealed and lubricated, replaceable antifriction guide and thrust bearings; and
 - Minimum 15,000 hours L-10 bearing life.
- 4) Inrush current shall not exceed six times motor full load current; and
- 5) At 100% load:

Submersible Pump Motors		
kW	Guaranteed Minimum Efficiency	Guaranteed Minimum Power Factor
3.7 through 7.4	80	82
7.5 through 36.9	85	82
37 through 75	87	82
Over 75	89	82

- 6) Winding Thermal Protection:
 - Submersible motors irrespective of kW rating, shall be fitted with PTC Thermostat.
- 7) Motor Seal Failure Moisture Detection:
 - Probes or sensors to detect moisture beyond seals;
 - Probe or sensor monitoring module for mounting in motor controller, suitable for operation from 24 volts AC supply; and
 - Monitoring module with control power transformer, probe test switch and test light, and two independent 24 volts AC contacts, one opening and one closing when the flux of moisture is detected.
- 8) Bearing Overtemperature Protection for Motors Larger than 100 kilowatts:
 - Sensor on lower bearing housing monitoring bearing temperature; and
 - Any monitoring relay necessary to provide 230 volts AC contact opening on bearing overtemperature.

- 9) Winding thermal protection, moisture detection, and bearing overtemperature specified above may be monitored by a single device providing two independent 230 volts AC contacts, one closing and one opening on malfunction.
- 10) Connecting Cables:
 - Two separate cables, one containing power and earthing conductors, and the other containing control and earthing conductors;
 - Each cable suitable for hard service, submersible duty with watertight seal where cable enters motor;
 - Length: 10 metres minimum;
 - Sized to meet applicable requirements of CP 5 and SS 551; and
 - Terminal boxes shall be factory sealed and rated for continuous submersion.
- (viii) Inclined Motors
 - 1) Motors suitable for operation only in horizontal position not acceptable.
 - 2) Bearings designed for thrust imposed by driven equipment and by motor rotor when motor is in inclined position.
 - 3) Lubrication system designed to provide adequate bearing lubrication when motor is in inclined position.
- (q) Testing
 - (i) Tests

Routine Tests on all motors in accordance with IEC 60034, plus no load power at rated voltage and polyphase, rated voltage measurement of locked rotor current. Test multispeed motors at all speeds.

 - 1) For energy efficient motors, test efficiency at 50, 75, and 100 percent of rated kilowatts:
 - In accordance with IEC 60034-12;
 - Power factor;
 - Speed;
 - Current at nameplate rating;
 - kW input; and
 - On motors of 100 kilowatt and smaller, provide a certified copy of a motor efficiency test report on an identical motor.
 - (ii) Test Report Forms
 - 1) Routine Tests shall be done in accordance with IEC 60034; and
 - 2) Efficiency and power factor by testing shall be done in accordance with IEC 60034.

G5.38.3 Execution

- (a) Installation
 - (i) In accordance with manufacturer's instructions and recommendations;
 - (ii) Align motor carefully and properly with driven equipment; and
 - (iii) Secure equipment to mounting surface with anchor bolts. Provide anchor bolts meeting manufacturer's recommendations and of sufficient size and number for the specified seismic conditions.
- (b) Field Inspection and Testing
 - (i) General: Inspection and testing on all motors;
 - (ii) Visual and Mechanical Inspection
 - 1) Proper electrical and earthing connections;
 - 2) Inspect each motor for any deviation from rated voltage, number of phases, winding connections, frequency etc;
 - 3) Check correct calibration of winding and bearing temperature detectors, for functional operation;
 - 4) Test for proper rotation prior to connect to the driven equipment;
 - 5) Shaft alignment;
 - 6) Blockage of ventilating air passageways;
 - 7) Operate motor and check for:
 - Excessive mechanical and electrical noise;
 - Overheating of motors, bearings and vibration behaviour for 4 hrs at full load and rated speed. The vibration limits used for shop testing with temporary support shall also be applicable to the field testing condition with proper rigid support structure below the machinery;
 - Motor operation shall be tested from LCS, LCP and SCADA for testing the status, and control functions;
 - Correct rotation;
 - Check vibration detectors, resistance temperature detectors, or motor inherent protectors for functionality and proper operation; and
 - Excessive vibration.
 - 8) Check operation of space heaters; and
 - 9) Earthing connection.

- (iii) Electrical Tests
 - 1) Insulation Resistance Tests:
 - Motors above 150 kW for 10-minute duration with resistances tabulated at 30 seconds, 1 minute, and 10 minutes; and
 - Motors 150 kW and less for 1-minute duration with resistances tabulated at 30 and 60 seconds.
 - Insulation resistance values equal to, or greater than, ohmic values established by manufacturers.
 - (iv) Calculate polarisation index ratios for motors above 150 kW. Investigate index ratios less than 1.5 for Class A insulation and 2.0 for Class B insulation.
 - (v) Insulation resistance test on insulated bearings in accordance with manufacturer's instructions.
 - (vi) Measure running current and voltage, and evaluate relative to load conditions and nameplate full-load amperes.
- (c) Manufacturer's Services
 - (i) See General Specification G2.25 – Manufacturers' Services, and G2.26 - Plant Testing and Commissioning.
 - (ii) Provide Manufacturer's Certificate of Proper Installation.

END OF SECTION